

Development and calibration of coarse-grained models of bacterial growth and bioproduction

Nabigha Mogharbel^{1,2}, Natale Scaramozzino^{1,2}, Delphine Ropers^{1,2}, Hidde de Jong^{1,2}

*For correspondence:
nabigha.mogharbel@inria.fr (NM)

¹INRIA, Université Grenoble Alpes research centre, Montbonnot, Saint Ismier Cedex, France; ²Université Grenoble Alpes, Laboratoire Interdisciplinaire de Physique (CNRS UMR 55888), Saint Martin d'Hères, France

Abstract Many valuable molecules are produced by bacteria, naturally or genetically modified, such as 1,3-propanediol (PDO). Coarse-grained models are promising for encapsulating important aspects of bacterial bioproductions such as efficiency and productivity. We have developed coarse-grained model for bio-production of 1,3-PDO in transformed *E.coli* grown in batch and calibrated the wild type model with data from literature. In addition, we proposed a feedback function for resource allocation between metabolic machinery and ribosomal machinery. The model was validated with experimental result of *E.coli* grown in batch on minimal medium with glucose. The model showed initial promising results and will be further developed and validated with future experiments with transformed *E.coli* which produces 1,3-PDO.

Introduction

Microorganisms, like bacteria, has been extensively used for the production of valuable biomolecules. An example of a valuable molecule is 1,3-propanediol (PDO), which is naturally produced by some microorganisms like *Klebsiella pneumoniae* or *Clostridium butyricum* from glycerol in anaerobic conditions. With the existing of biodiesel industry, glycerol is synthesized in bulk as a by-product (Agrawal *et al.*, 2023). Several studies focused on the commercial bio-synthesis of 1,3-PDO in metabolically engineered *E.coli* from glucose or glycerol either via genetic engineering on the chromosome level or by using plasmids (Frazão *et al.*, 2019; Izard *et al.*, 2015; Wong and Jantama, 2022; Liang *et al.*, 2011). However, the optimization of its production efficiency is still a challenge due to the complex relations between bacterial metabolism, protein synthesis, and growth (Agrawal *et al.*, 2023).

Modeling microorganism's metabolism is key aspect in biotechnology, since it allows a mathematical representation of the metabolism and thus opening the window to analyze, optimize and predict bacterial behaviour. Two main types of models exists, genome-scale models (GSM) and coarse-grained (CG) models. GSMs are comprehensive mathematical representation of the entire metabolic network of the organism. They integrates genomic and biochemical data to evaluate reactions' fluxes. However, they lack definitions like enzyme kinetics and their simulations are computationally expensive (Schroeder *et al.*, 2024). On the other hand, CG models simplify the metabolism by grouping cellular components into functional classes such as metabolic machinery, ribosomal and protein synthesis machinery, and housekeeping proteins. These definitions allow integrating enzyme kinetics as well (Doan *et al.*, 2022).

While GSMs provide detailed metabolic mapping, CG models are powerful at capturing the essential regulatory principles and trade-offs of microbial growth strategies. Due to these advantages, and their lower computational cost, they were shown to be promising in the applications of biotechnology (Schroeder et al., 2024; Yegorov et al., 2019).

Scott et al. (2010) results showed that the proportion of ribosomal proteins changes according to the growth rate of bacteria grown on different carbon sources. Regulatory feedback mechanisms like ppGpp regulate the synthesis of ribosomal proteins and metabolic proteins based on the amount of precursors available in the cell. This helps the bacteria to keep the energy supply sufficient. In addition, some carbon sources like acetate needs additional metabolic pathways to be activated before entering the carbon cycle, imposing more demand on metabolic proteins (Scott et al., 2014). In the case of CG models, modeling this feedback is known as resource allocation strategy. Several several proposed relationship have been defined previously due to its effect in predicting the growth rate, biomass yield and substrate uptake rate (Scott et al., 2014; Giordano et al., 2016; Yegorov et al., 2019; Yabo et al., 2022).

This research focused on defining a CG model for bacteria grown in batch and produce 1,3-PDO. We started by defining a model inspired by pre-existing CG models in literature and we proposed a definition of resource allocation function. We calibrated the wild-type model using data from literature on *E.coli* grown on different carbon sources in batch. We validated the predictions of the model by experimentally growing *E.coli* on minimal medium with glucose and analyzed the model's predictions in terms of productivity (growth rate) and efficiency (biomass yield).

Results

Coarse-grained model

Our coarse-grained model is based on previously established models which described the bacterial growth by a small number of macroreactions. It also makes several assumptions, first: protein degradation is ignored, second: we group proteins together into several classes rather than considering proteins individually, and third: we describe a population of bacteria as identical cells (Scott et al., 2010; Giordano et al., 2016; Yegorov et al., 2019; Yabo et al., 2022; Baldazzi et al., 2023).

We defined the cell as a system composed of external substrate S , precursor metabolites P , house-keeping proteins Q , gene expression machinery R , metabolic machinery M , heterologous protein W , and metabolite of interest X which represents the mass of these components. The mass flow between these different components were represented by V_M for the rate transformation of S into P , V_R for the rate of protein synthesis (Q , M and R machinery) from P , and V_X for rate of X production from P .

The proportion of V_R allocated between Q and the remaining non-housekeeping proteins were defined by parameter $1-\gamma(t)$ and $\gamma(t)$ where $\gamma(t) \in [0, 1]$. The proportions allocated to non-housekeeping proteins were further divided between R , M , W using $\alpha(t) \in [0, 1]$ and $\chi(t) \in [0, 1]$.

Following previous work, the mass quantities S , P , R , M , Q , W , X can be defined in terms of mass per cell volume or external volume (for substrate) to obtains the concentration quantities s , p , r , m , q , w , x . Similarly, the masses flow rates V_M , V_R , V_X can be defined as the masses flow rates per unit volume to obtain v_M , v_R , v_X .

Using these definitions, our system can be written as system of ordinary differential equations (ODE)s as shown in Figure 1 which describe the rate of change of concentration quantities over time $t \in (0, T)$ with initial concentrations $s_0, p_0, r_0, m_0, q_0, w_0, x_0$ at $t = 0$. The mass flow per unit volume can be expressed as a functions of the available concentration quantities following Michaelis-Menten kinetics, where k_M, k_R, k_X are the maximal reaction rates and K_M, K_R, K_X are the half saturation constants. The detailed model derivation is described in the *materials and methods* section.

The Michales-Menten relations tells us that the synthesis rate of precursors is proportional to the rate of metabolic machinery while the synthesis rate of macromolecules M , R , Q , W is proportional to the gene expression machinery. In addition, the synthesis rate of precursors depends on the

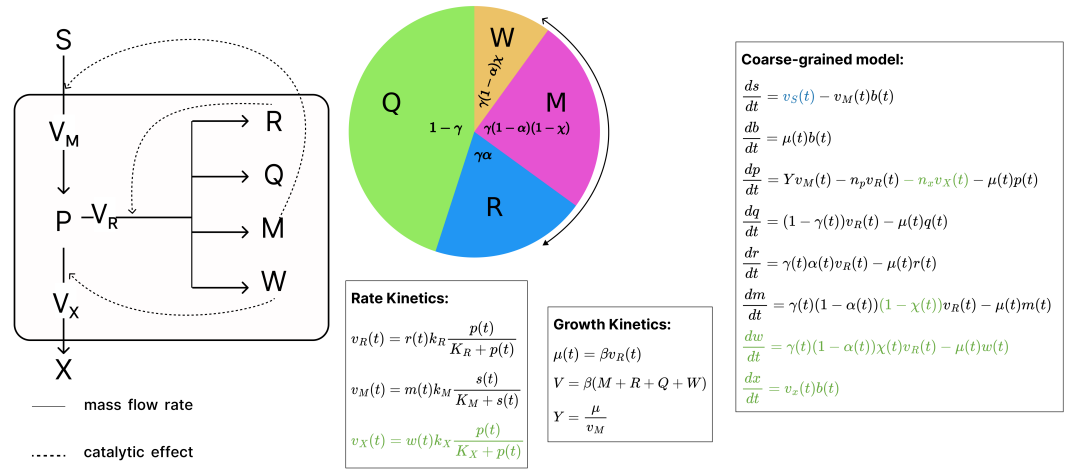


Figure 1. Coarse-grained model for 1,3-PDO production in *E. coli*. External substrate S is supplied at rate V_S and consumed at rate V_M to make precursor metabolites P through metabolic machinery M . P is then used for the synthesis of macromolecules M , gene expression machinery R , housekeeping machinery Q , heterologous enzymes W at different proportions of protein synthesis rate V_R and 1,3-PDO synthesis X at rate V_X . The different proportion of rate V_R to synthesize protein macromolecules are defined by γ , α and χ . Solid arrows represent the mass flow while dashed arrows represent catalytic effects such as regulatory mechanisms, metabolism and 1,3-PDO synthesis. The equations written in black represent the model for wild type bacteria grown in batch mode where substrate uptake rate v_s is 0. The blue part of the equations represents a constant feeding conditions, like a bioreactor. The green part of the equations represents the production of heterologous proteins and 1,3-PDO metabolite.

constant k_M which reflects the quality of the nutrient in the medium and the efficiency of the reaction; higher k_M means richer medium and more efficient reaction. The constant K_M describes the substrate concentration where the reaction rate is half of maximal synthesis rate $v_{M,max}$.

As we have mentioned previously, $\gamma(t)$ controls the resource allocation of protein synthesis between Q and other protein classes. Although we have expressed it as a function over time, it has been shown that housekeeping protein machinery constitute a constant proportion of bacterial proteome which is approximated to be $\gamma(t) = 0.45$ (Scott et al., 2014). $\alpha(t)$ controls the resource allocation of v_R for R , M , W but in contrary to $\gamma(t)$, its value is not constant but depend on the current state of the bacteria. Biologically, α resembles a feedback mechanism which modifies the resource allocation strategy based on the available precursors. It has been shown previously that this function resembles a Hill function (Scott et al., 2014; Giordano et al., 2016). Following the previous results, we suggested the following definition (Eq: 1) where Θ is a parameter we will approximate later:

$$\alpha(t) = \frac{p^2}{p^2 + \Theta^2} \quad (1)$$

Model calibration

We start calibrating our wild type model for bacteria grown in batch mode on minimal medium with glucose defined by the system of ODEs in black in Figure 1. For estimating the parameters, we will ignore that α is a function of p .

1- Estimating total protein concentration $1/\beta$

Total protein concentration ($\frac{1}{\beta}$) was reported by Basan et al. (2015) and reproduced by Baldazzi et al. (2023). We took average value of 0.7 g gDW^{-1} which we transformed into $0.0212 \text{ mmol gDW}^{-1}$ following Milo (2013) estimation (Eq: 2) where $m_{tot} = 0.7$ is our average protein concentration,

$l_p = 300$ is the average number of amino acids in one protein, $m_{aa} = 110$ is the average mass of one amino acid in Dalton, $C_{gDa} = 1.66054 \cdot 10^{-24}$ is the conversion factor from Dalton to gram, $NA = 6.02214 \cdot 10^{23}$ is Avogadro's constant which defines the number of molecules per mole:

$$\frac{1}{\beta} = \frac{m_{tot}[g/gDW]}{l_p [aa/protein] \cdot m_{aa} [Da/aa] \cdot C_{gDa} [g/Da]} \cdot \frac{1}{NA} [mol/molecule] \cdot 10^3 [mmol/mol]. \quad (2)$$

2- Estimating Resource Allocation Parameters γ and α

Resource allocation parameters γ and α describe the proportions of protein synthesis rate allocated to non-housekeeping proteins and ribosomal proteins respectively. **Scott et al. (2010)** reported measurements of $\Phi_R = \frac{R_{protein}}{total\ protein}$ as a relation with growth rates. Using these results, we found Φ_R to be 0.195 at growth rate 0.79 h^{-1} ; the growth rate of our bacteria strain on minimal medium (M9) with glucose (**Pavlou et al., 2025**). He also defined the relation $\Phi_Q = 1 - \Phi_R^{max} = 0.45$ which is constant value regardless of the growth rate. To adapt these values to our model, we calculated the value of γ to be 0.55 using equation 3 then calculated $\alpha = 0.355$ using the equation 4:

$$\Phi_Q = 1 - \gamma. \quad (3)$$

$$\Phi_R = \gamma \alpha. \quad (4)$$

3- Estimating biomass yield Y

Biomass yield Y is defined as the fraction of substrate molecules assimilated into biomass. The yield can be defined as g of biomass obtained per g of substrate. We took the biomass yield (Y) reported by **Gerosa et al. (2015)** which were measured to be 0.37 g/g . We converted the gram of substrate into mmol by dividing over molecular weight which gave us $0.0667 \text{ gDW mmol}^{-1}$.

4- Estimating Precursors Concentrations p

An estimate of the total metabolite concentrations were reported by **Gerosa et al. (2015)**. It can be used to compare with our estimate of p during exponential growth. Metabolite concentrations are measured to be on the order of $100 \mu\text{M gDW}^{-1}$.

5- Deriving protein concentrations q , r and m

We consider the population during exponential growth, which means that intracellular quantities are at steady state (indicated by $*$). This leads to the following relations for the protein concentrations:

$$\begin{aligned} q^* &= (1 - \gamma) \frac{1}{\beta}, \\ r^* &= \alpha \gamma \frac{1}{\beta}, \\ m^* &= (1 - \alpha) \gamma \frac{1}{\beta}. \end{aligned} \quad (5)$$

The values of q^* , r^* , m^* can be determined from the measured values for α , γ .

6- Deriving substrate uptake rate v_M

Using equation 6 and the values of Y and μ^* we estimated above, we were able to calculate the uptake rate of glucose to be $11.2515 \text{ [mmol gDW}^{-1} \text{ h}^{-1}]$:

$$Y = \frac{\mu^*}{v_M^*}. \quad (6)$$

7- Deriving half saturation constants K_M and K_R

The value of K_M is known to be low for most substrates in the μM range (**Boulineau et al., 2013**). We estimate the value of K_M as $5 \mu\text{M}$, that is, $5 \mu\text{ mol L}^{-1}$. K_R is expected to have the same order of magnitude as the steady-state precursor concentration during growth on glucose (**Bennett et al., 2009**). We therefore equate this value to the estimate of p^* mentioned above. The definition of the protein synthesis rate can then be used to estimate k_R .

8- Deriving maximum rate constants k_M and k_R

During exponential growth in batch and as mentioned in the previous section, K_M values are estimated to be in the low μM range, thus $K_M \ll s$ which give us the following approximation:

$$v_m^* \approx k_M m^*. \quad (7)$$

We used the definition of m^* from equation 5 and derived the maximum rate constants of uptake rate k_M to be 1494.16 h^{-1} . For deriving maximum rate constants for protein synthesis k_R , we used the value of K_R derived from previous section which can be approximated into p^* which gave us the expression 8 and allowed us to find k_R value equals to 7.69 h^{-1} .

$$\mu^* = \beta v_R = \beta r^* k_R \frac{p^*}{2p^*}. \quad (8)$$

9- Estimating Stoichiometry Constant n_p

Estimating stoichiometry constant n_p represents the number of substrate molecules needed to make 1 protein molecule. In this work, we choose a value that gave the best performance of the predictions which we found to be 50. Table 1 summarizes the parameters above:

Table 1. Summary of model parameters.

Parameter	Value	Unit	Parameter	Value	Unit
$1/\beta$	0.02	mmol gDW ⁻¹	γ	0.55	-
α	0.36		Y	0.07	gDW mmol ⁻¹
n_p	50		K_M	0.005	mmol L ⁻¹
k_M	1494.16	h ⁻¹	K_R	0.1	mmol gDW ⁻¹
k_R	7.69	h ⁻¹	p	0.1	mmol gDW ⁻¹
r	$\gamma \alpha \frac{1}{\beta}$	mmol gDW ⁻¹	m	$\gamma (1 - \alpha) \frac{1}{\beta}$	mmol gDW ⁻¹
q	$(1 - \gamma) \frac{1}{\beta}$	mmol gDW ⁻¹			

10- Parameters of acetate

We did the same estimations above and many parameters were the same. The only parameters we changed were and we found values only the parameters of $\alpha = 0.18$, $Y = 0.03 \text{ gDW mmol}^{-1}$, $n_p = 60$, $k_M = 1258.21 \text{ h}^{-1}$, $k_R = 4 \text{ h}^{-1}$.

11- Wild type model simulation

We simulated the model using the estimated parameters summarized in the previous section on both glucose and acetate. We choose s_0 , the initial substrate concentration of glucose and acetate to be 1 g L^{-1} , which we converted to mmol L^{-1} by dividing the mass in gram over the molecular weight of substrate and we obtained $5.551 \text{ mmol L}^{-1}$ for glucose and $16.937 \text{ mmol L}^{-1}$ for acetate. The initial biomass b_0 was fixed to $400 \mu\text{gDW L}^{-1}$. The values for α were fixed as well and we simulated the system a couple of hours after the substrate depletion. The results of the simulations are represented by solid lines in Figure 2A and B. The growth rate drops rapidly during the first phase of the simulation, then stay fixed over the time before it starts decreasing until 0. The constant part of the growth curve resembles the growth rate in exponential phase where growth rate is constant, thus we took the growth rate value from this constant part of the curve. The growth rate was 0.65 h^{-1} on glucose and 0.183 h^{-1} on acetate. The maximum biomass was 0.35 gDW L^{-1} in the case of glucose and 0.275 gDW L^{-1} in the case of acetate.

The concentration of p^* during exponential phase was $0.078 \text{ mmol gDW}^{-1}$ on glucose and $0.087 \text{ mmol gDW}^{-1}$ on acetate represented by a dot in the plot.

To find the yield using the simulation results (and later the experimental results), we found calculated the change of b from time 0 until final simulation time, which will represents the total increase

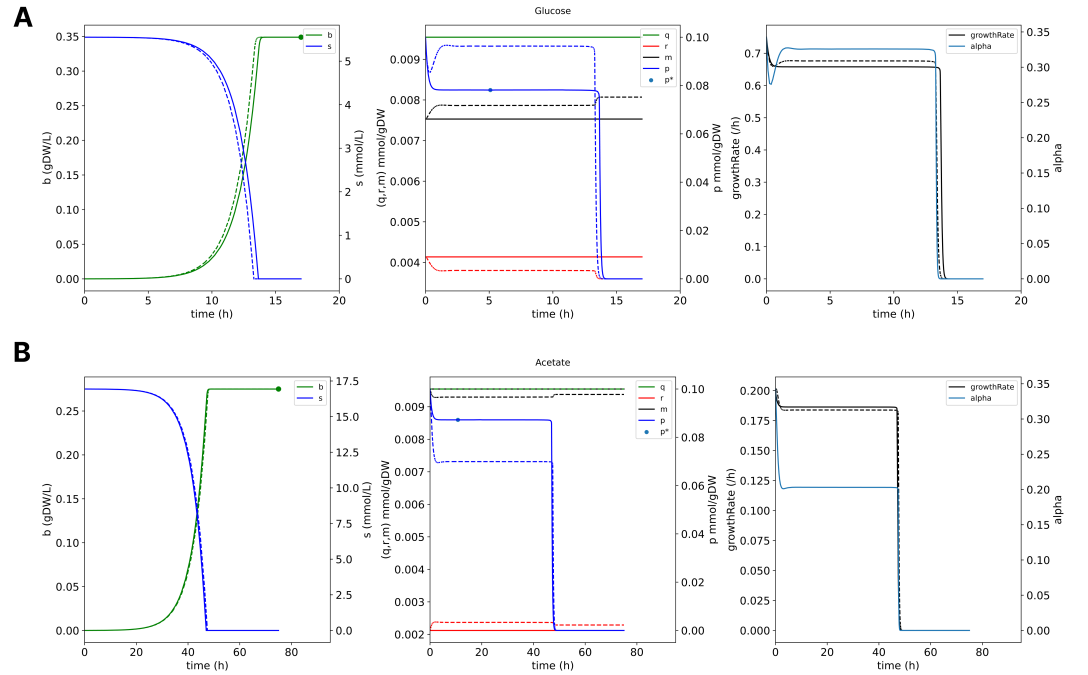


Figure 2. Model simulation with constant resource allocation and resource allocation expressed as a function of precursors. The solid lines are the predictions obtained with the value of α fixed to the observed value during exponential growth. The dashed lines are the predictions obtained when α is defined as a function of the precursors (Eq. 1) and fitted to the data. A. Simulation of growth on glucose. B. Simulation of growth on acetate

of biomass of bacteria grown on the minimal medium with glucose or acetate. Then, we divide it by the initial concentration of substrate to find number of mmol of substrate assimilated into each gDW of the biomass. This is summarized by equation 9:

$$Y = \frac{b_f - b_i}{s_i} \quad (9)$$

Applying the above equation gave us a yield of 0.063 gDW mmol⁻¹ on glucose and 0.0162 gDW mmol⁻¹ on acetate.

The uptake rate v_M was calculated to be 10.47 mmol gDW⁻¹ h⁻¹ of glucose and 11.47 mmol gDW⁻¹ h⁻¹ of acetate using equation 6.

We then fitted the proposed function of α (Eq. 1) using the concentration of precursors at steady state p^* (Figure 2) and $\alpha_{glucose}$ and $\alpha_{acetate}$ to estimate the parameter Θ in both conditions. The result of fitting showed Θ is equal to 0.1385. The result of the function is shown in figure 3

Since α isn't static anymore, its value changed from the optimal over time with respect to precursors concentration leading to variations in the concentrations of r and m over time. When the substrate concentration starts depleting, the growth rate starts dropping until it reaches 0, where the biomass will remain constant. In the case of glucose, our resource allocation feedback α was slightly lower than to $\alpha_{glucose}$ (0.32) leading to slightly higher m and lower r concentrations while it was the opposite in the case of acetate $\alpha = 0.20$. The growth rate was 0.68 h⁻¹ on glucose and 0.18 h⁻¹ on acetate. The yield was 0.062 gDW mmol⁻¹ on glucose and 0.0162 gDW mmol⁻¹ on acetate while the uptake rate stayed the same. The concentration of p^* during exponential phase was 0.1 mmol gDW⁻¹ on glucose and 0.075 mmol gDW⁻¹ on acetate.

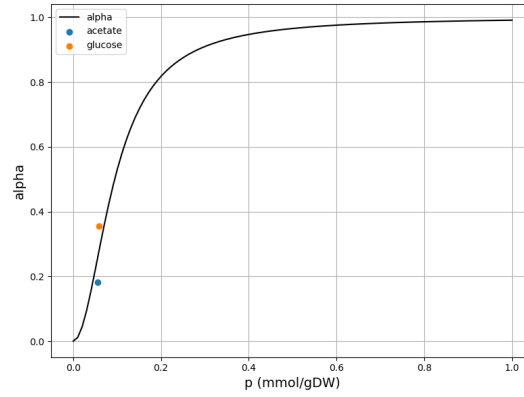


Figure 3. Resource allocation function α . the function was fitted against predicted concentrations of precursors at steady state p^* on both acetate and glucose. The figure shows the fitted function over a range of p concentrations $[0,1]$ mmol gDW⁻¹

Experimental Quantification of growth of *E.coli* on glucose

To measure the growth rate and the glucose uptake of BW25113 $\Delta fhuA rpsB-gfpmut2 argG-mScarlet-$ / on glucose, we have grown the bacteria in M9 with 0.1% glucose in a 96 wells microplate reader for 24 hours. The bacteria biomass were measured with an 600nm light (absorbance). Samples were taken from random wells over time, in duplicates, for glucose quantification.

Figure 4A shows the mean of raw and cleaned data of absorbance and fluorescence measurements over time for 5 non-sampled wells. To calculate the maximal growth rate, we fitted the absorbance values in log scale in the exponential phase where the growth rate represents the slope of the fitted line. The maximal biomass absorbance was calculated by finding the mean of the fitted blue line in Figure 4B. The mean growth rate of the 5 wells was measured to be 0.795 h⁻¹ Figure 4C.

We used 2.5 OD absorbance⁻¹ as a conversion constant from absorbance to OD and 0.49 gDW OD⁻¹ L⁻¹ to convert absorbance to gDW (Izard et al., 2015). We started the experiment with initial biomass (b_0) equals to 0.00049 gDW L⁻¹ and ended with mean maximal biomass (b_f) of 0.494 gDW L⁻¹. Our glucose quantification results shows initial glucose concentration was 1.45 g L⁻¹ which is equal to 8.05 mmol L⁻¹. We calculated the yield Y to be 0.061 gDW mmol⁻¹ using equation 9. The substrate uptake rate v_M was calculated to be 13.05 mmol gDW⁻¹ h⁻¹ using equation 6.

Model Predictions With Experimental Results

To evaluate the predictive capability of our calibrated coarse-grained wild type model, we compared the numerical simulations against kinetic data obtained from *E.coli* batch cultures grown on glucose. We started the simulations with initial conditions like to the experimental conditions. We used close to 0.00049 gDW L⁻¹ for b_0 and the initial concentration of glucose 8.05 mmol L⁻¹ (s_0). The system was simulated for the same duration of the microplate reader experiment (20 hours). We compared the model predictions using fixed value of α and our resource allocation function of α with the experimental conditions. The results are shown in Figure 5. The model's prediction of final biomass was 0.5 gDW L⁻¹, similar to experimental final biomass. The calculated yield was 0.06 and the uptake rate of glucose was 10.47 mmol gDW⁻¹ L⁻¹ for fixed α and 10.47 mmol gDW⁻¹ L⁻¹ for dynamic α . The growth rate was 0.65 h⁻¹ and 0.67 h⁻¹ respectively.

Discussion

In this study, we have developed a coarse-grained model for bio-production of 1,3-PDO in transformed *E.coli* with plasmid. We have calibrated the wild type model with available data from literature and proposed a resource allocation function which regulates proportion of synthesis of ribosomal machinery based on available precursors inside the cell. The model was validated against experimental result of *E.coli* grown on minimal medium with glucose.

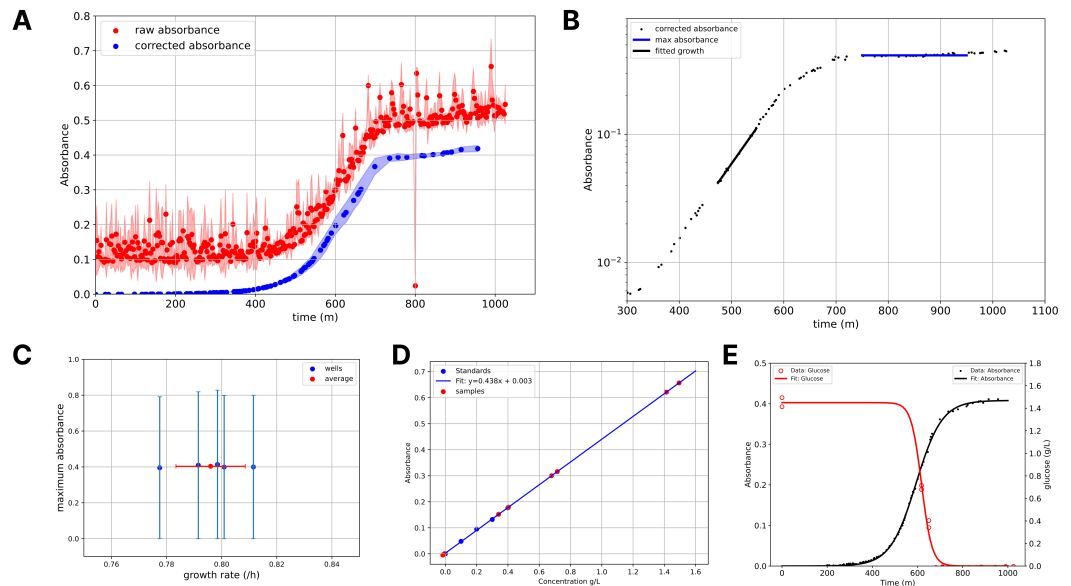


Figure 4. Growth kinetics of *E. coli* in M9 0.1% glucose. A. Raw and corrected absorbance measurements. B. Fitting of absorbance during exponential phase to calculate growth rate and fitting of absorbance level in stationary phase, for a single well C. Calculated growth rate and maximum absorbance for 5 wells. D. Fitting of glucose standard curve and quantification of glucose for samples. E. Fitting of absorbance and glucose concentration over time.

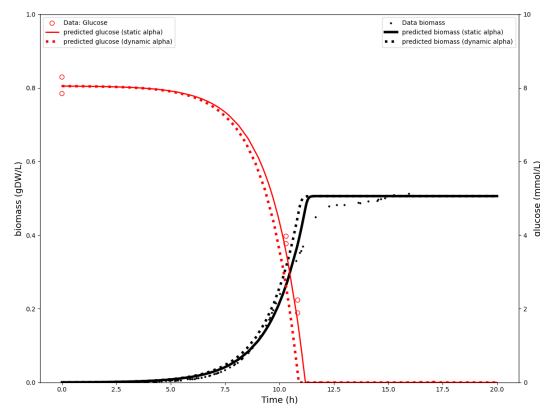


Figure 5. Model predictions compared with experimental Data. Experimental values are represented by dots, the solid line is the predictions obtained with the value of α fixed to the observed value during exponential growth and the dashed line is the predictions obtained when α is defined as a function of the precursors (Eq. 1)

To verify that 0.7 gram of protein per gDW of cell resembles a reasonable number, dividing average protein mass of one *E.coli* cell ($3 \cdot 10^{-13}$ gDW $\cdot$ 0.7 g gDW⁻¹) over average protein mass in g, we obtain gDW⁻¹ ($3.8 \cdot 10^6$ as a number of proteins per cell, which is close to Milo's estimated number of proteins in *E.coli* (Milo, 2013). Similarly, our calibrated uptake rate of glucose and acetate matches Gerosa et al. (2015) measured uptake rates. We have initially thought of the parameter n_p to be equal to 300, however, our model performance was far from accurate. Although we have settled on the values 50 and 60 for glucose and acetate but our choice is still open for discussion and optimization.

The models predictions using fixed value of α showed similar yield but lower growth rate of glucose and slightly higher for acetate.

Our proposed resource allocation function showed close values to what we found in literature showing a sub-optimal concentrations of r and m in the case of glucose and acetate. Our fitted function of resource allocation gave a concentration of precursors in steady state close to what we found in literature (0.1) in the case of glucose, while it was much lower in the case of acetate. The growth rate remained close to what we observed with fixing α .

Our calculated growth rate of *E.coli* in minimal medium with glucose 0.79 h⁻¹ and the calculated yield matched previously measured growth rate and data from literature (Pavlou et al., 2025; Gerosa et al., 2015). However, the glucose uptake rate was higher compared to literature which might be due to low number of sampling or a problem in our measurements (Gerosa et al., 2015). The model's prediction compared to experimental data showed lower glucose uptake rate and lower growth rate while the final biomass and yield were the same. This is an example of how changing the growth rate of bacteria affects substrate uptake rate while having the same final biomass yield at the end.

Overall, the model showed initial promising results, some parameters needs further optimization such as defining resource allocation in term of r and p similar to previous work (Giordano et al., 2016). For the case of bioproduction of 1,3-PDO, it is known from literature that synthesis of biomolecules results in lower biomass yield and slower growth rate due to the competition on carbon between cellular natural metabolism and synthesis of heterologous proteins, byproducts accumulation, and energy vs yield trade-off (?). This project will continue during my PhD to perform quantification experiments on carbon source and 1,3-PDO production, optimization of the plasmid design and metabolic network of 1,3-PDO, and further development and adaptation of the model to resembles bacterial growth in bio-reactor which are closer to actual bioproduction environment.

Methods and Materials

E.coli strain

We used two strains derived from *E.coli* K-12 wild-type strain BW25113; BW25113 $\Delta fhuA$ *rpsB-gfpmut2 argG-mScarlet-I* as described in a previous study (Pavlou et al., 2025). GFPmut2 was fused to the C-terminus of *rpsB* which encodes the S2 ribosomal subunit. mScarlet fused to the C-terminus of *argG* which encodes the enzyme involved in the biosynthesis of L-arginine.

Minimal medium (M9)

We used M9 minimal medium for bacterial growth composed of 0.1 mM CaCl₂, 2 mM MgSO₄, 5mg/L Vitamin B1, 47.8mM Na₂HPO₄, 22 mM KH₂PO₄, 8.5 mM NaCl, 18.7 mM NH₄Cl, 28 μ M Na₂EDTA, 15.6 μ M ZnSO₄, 1.4 μ M CoCl₂, 5 μ M MnCl₂, 1.6 μ M Na₂MoO₄, 6.5 μ M FeSO₂, 16 μ M H₃BO₃, 1.2 μ M CuSO₄. To make M9 0.1% glucose, we add 150 μ L of glucose 20% (20 g/100 mL) per 30 mL M9 (Pavlou et al., 2025).

Microplate reader and sampling

We started the experiment from 1 colony of BW25113 $\Delta fhuA$ *rpsB-gfpmut2 argG-mScarlet-I* on agar-LB plate and incubated it in minimal medium with 0.1% glucose overnight to make a pre-culture.

Pre-cultures were then washed by centrifuging on 11G, 4°C for 3 minutes then re-suspended in M9. Bacterial density was measured by mean of a spectrometer at 600 nm wavelength. The needed volume of each re-suspended bacteria in the minimal medium + 0.1% carbon source is calculated using this formula $V_{M9} * OD_f / OD_i$ to ensure optical density (OD_f) is 0.00. 150µL of M9 0.1% carbon source with bacteria is transferred to 96-wells Greiner Flat Bottom Black Polystyrol microplate. 2mm glass beads are added to each well to ensure oxygenation. M9 0.1% carbon source without bacteria was added to some wells for background correction. Microplate is measured for 24 hours using infinite 200Pro microplate reader running on Tecan i-control 1.10.4 software. To take samples over time for carbon source quantification, we transferred 150µL from microplate well and centrifuged at 11G, 4°C for 3 minutes. Supernatants were transferred into a transparent 96-wells microplate and stored at -20°C.

Growth-rate calculation

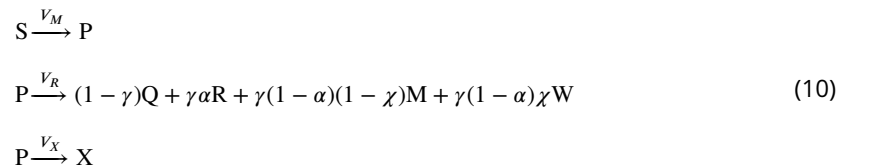
The microplate reader data were first treated with WellInverter; a software tool designed specifically for this type of experiments (Martin et al., 2019). The absorbance curve is manually cleaned from outliers due to the presence of glass beads in the wells and corrected to a the absorbance background using the average absorbance value of a negative control well (well with M9 without bacteria). The cleaned data were then used to calculate the growth rate by fitting a straight line to the natural logarithm of the absorbance over time in exponential phase. The fitting and calculations were done using Python 3.12.

Glucose quantification

We used Megazyme Kit to quantify the concentration of glucose in the medium over time as described by the manufacturer's protocol (Megazyme, 2023). In brief, the samples were diluted such that the maximum concentration will be within the detection range of the kit. Standards samples were created by diluting standard solution (0.4 mg/mL glucose) with different dilution factors. Reagents and enzyme were added as described and optical density is measured before and after the addition of enzymes (ΔOD_{340nm}). Standard curve is done by fitting the ΔOD_{340} with the concentration of the standards (Figure 4D). The concentration of samples were determined through the standard curve multiplied by the dilution factor of the sample. The curve fitting is done using Python.

Model definition

Our model follows previously established coarse-grained resource allocation models (Giordano et al., 2016; Yegorov et al., 2019; Yabo et al., 2022). S represents the substrate outside the cell and P represents the precursors inside the cell. We partitioned the proteome into four main categories of proteins: house-keeping proteins Q, protein synthesis machinery (ribosomes) R, metabolic machinery M and our 4 heterologous proteins W. The heterologous proteins W will synthesize heterologous metabolite X, which is 1,3-PDO in our case. The overall allocation process is represented by these macroreactions:



We define Y , the biomass yield constant, as the fraction of substrate molecules assimilated into biomass and n_p, n_x , stoichiometry constant, as approximate number of precursor molecules required for synthesizing a single molecule. Generalizing the model as described by the mentioned

previous models give this dynamical system:

$$\begin{aligned}
\frac{dS}{dt} &= V_S(t) - V_M(t), \quad S(0) = S_0 \\
\frac{dP}{dt} &= YV_M(t) - n_p V_R(t) - n_x V_X(t), \quad P(0) = P_0 \\
\frac{dQ}{dt} &= (1 - \gamma(t))V_R(t), \quad Q(0) = Q_0 \\
\frac{dR}{dt} &= \gamma(t)\alpha(t)V_R(t), \quad R(0) = R_0 \\
\frac{dM}{dt} &= \gamma(t)(1 - \alpha(t))(1 - \chi(t))V_R(t), \quad M(0) = M_0 \\
\frac{dW}{dt} &= \gamma(t)(1 - \alpha(t))\chi(t)V_R(t), \quad W(0) = W_0 \\
\frac{dX}{dt} &= V_X(t), \quad X(0) = X_0
\end{aligned} \tag{11}$$

The variables $V_S(t)$, $V_M(t)$, $V_R(t)$, $V_X(t)$ represents the rates of external substrate, production of precursors, synthesis of proteins, and synthesis of propanediol, respectively [mmol h⁻¹].

Similarly to previous models, we assume that cellular density is constant (*Giordano et al., 2016; Yegorov et al., 2019*), thus we can define total biomass $B(t)$ and total volume $\mathcal{V}(t)$ like below where β is the inverse of total protein concentration [gDW mmol⁻¹]:

$$B(t) = M(t) + R(t) + Q(t) + W(t) \quad [\text{mmol}] \quad \mathcal{V}(t) = \beta(M(t) + R(t) + Q(t) + W(t)) \quad [\text{gDW}] \tag{12}$$

Growth rate $\mu(t)$ can be written as the relative increase of volume:

$$\mu(t) = \frac{1}{\mathcal{V}(t)} \frac{d\mathcal{V}}{dt} \tag{13}$$

To express the system in terms of concentrations, i.e. mass quantities per unit population volume $\mathcal{V}(t)$ or per fixed constant external volume $\mathcal{V}_{ext}$ [L], we introduce the following variables:

$$\begin{aligned}
p(t) &= \frac{P(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1}] & q(t) &= \frac{Q(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1}] \\
m(t) &= \frac{M(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1}] & r(t) &= \frac{R(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1}] \\
w(t) &= \frac{W(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1}] & s(t) &= \frac{S(t)}{\mathcal{V}_{ext}} \quad [\text{mmol L}^{-1}] \\
x(t) &= \frac{X(t)}{\mathcal{V}_{ext}} \quad [\text{mmol L}^{-1}] \\
v_R(t) &= \frac{V_R(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1} \text{ h}^{-1}] & v_X(t) &= \frac{V_X(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1} \text{ h}^{-1}] \\
v_M(t) &= \frac{V_M(t)}{\mathcal{V}(t)} \quad [\text{mmol gDW}^{-1} \text{ h}^{-1}] & v_S(t) &= \frac{V_S(t)}{\mathcal{V}_{ext}} \quad [\text{mmol L}^{-1} \text{ h}^{-1}]
\end{aligned} \tag{14}$$

Using the growth rate definition (Eq. 13), we can rewrite the system of Eq. 11 as Eq. 15 using variables from Eq. 14:

$$\begin{aligned}
\frac{ds}{dt} &= v_S(t) - b(t)v_M(t), \quad s(0) = s_0, \\
\frac{db}{dt} &= \mu(t)b(t), \quad b(0) = b_0, \\
\frac{dp}{dt} &= Yv_M(t) - n_p v_R(t) - n_x v_X(t) - \mu(t)p(t), \quad p(0) = p_0, \\
\frac{dq}{dt} &= (1 - \gamma(t))v_R(t) - \mu(t)q(t), \quad q(0) = q_0, \\
\frac{dr}{dt} &= \gamma(t)\alpha(t)v_R(t) - \mu(t)r(t), \quad r(0) = r_0, \\
\frac{dm}{dt} &= \gamma(t)(1 - \alpha(t))(1 - \chi(t))v_R(t) - \mu(t)m(t), \quad m(0) = m_0, \\
\frac{dw}{dt} &= \gamma(t)(1 - \alpha(t))\chi(t)v_R(t) - \mu(t)w(t), \quad w(0) = w_0, \\
\frac{dx}{dt} &= v_x(t)b(t), \quad x(0) = x_0.
\end{aligned} \tag{15}$$

Following previous work, we write the formulas of v_M, v_R, v_X where k_M, k_R, k_X are rate constants [h^{-1}] and K_M, K_R, K_X are half saturation [mmol gDW^{-1}]:

$$\begin{aligned}
v_M(t) &= m(t)k_M \frac{s(t)}{K_M + s(t)} \\
v_R(t) &= r(t)k_R \frac{p(t)}{K_R + p(t)} \\
v_X(t) &= w(t)k_X \frac{p(t)}{K_X + p(t)}
\end{aligned} \tag{16}$$

By using equations 12 11 and 16, we can rewrite growth law to be:

$$\mu(t) = \beta v_R(t) \tag{17}$$

Code Availability

All the code to analyse and reproduce the figures is available through this link:

<https://gitlab.inria.fr/nmogharb/coarsegrainedmodelsforbioproduction>

Acknowledgments

Thanks for my supervisors, Hidde de Jong, Delphine Ropers and Natale Scaramozzino for their support during my internship with all the technical issues we have faced. Thanks for Hidde for running many pre-cultures during the night (10:00PM) or the weekends so I can save time. Thanks to Delphine for her course in my Master 1 degree which inspired me to work in this field and eventually join her team. Thanks for my teams MICROCOSME and BIOP for their assistance and spirit. Thanks to Rand for giving me a crash course in optimal control theory within 2 hours.

Thanks for my friends in Grenoble for their mental and physical support (Ruella, Hadi, Nidhi, Sheelsen and Loick) and for offering me a place to sleep in when I needed it. Thanks to my beloved city Grenoble that made me meet amazing people and experience all of that. Most importantly, thanks for my family (Ghina, Ahmad, Rassil, Yazan) for everything they helped me with even from distance, and simply, just for existing in my life. Without everyone I mentioned here, I wouldn't be able to make it.

Finally, thanks for the Graduate School - Planned Health for funding this internship.

References

- Agrawal D**, Budakoti M, Kumar V. Strategies and tools for the biotechnological valorization of glycerol to 1, 3-propanediol: Challenges, recent advancements and future outlook. *Biotechnology Advances*. 2023; 66:108177. <https://doi.org/10.1016/j.biotechadv.2023.108177>, doi: 10.1016/j.biotechadv.2023.108177.
- Baldazzi V**, Ropers D, Gouzé JL, Gedeon T, De Jong H. Resource allocation accounts for the large variability of rate-yield phenotypes across bacterial strains. *eLife*. 2023; 12:e79815. <https://elifesciences.org/articles/79815>, doi: 10.7554/eLife.79815.
- Basan M**, Zhu M, Dai X, Warren M, Sévin D, Wang YP, Hwa T. Inflating bacterial cells by increased protein synthesis. *Molecular Systems Biology*. 2015; 11(10):836. <https://doi.org/10.15252/msb.20156641>, doi: 10.15252/msb.20156641.
- Bennett BD**, Kimball EH, Gao M, Osterhout R, Van Dien SJ, Rabinowitz JD. Absolute metabolite concentrations and implied enzyme active site occupancy in *Escherichia coli*. *Nature Chemical Biology*. 2009; 5(8):593–599. <https://www.nature.com/articles/nchembio.186>, doi: 10.1038/nchembio.186.
- Boulineau S**, Tostevin F, Kiviet DJ, ten Wolde PR, Nghe P, Tans SJ. Single-cell dynamics reveals sustained growth during diauxic shifts. *PLoS ONE*. 2013; 8(4):e61686. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0061686>, doi: 10.1371/journal.pone.0061686.
- Doan DT**, Hoang MD, Heins AL, Kremling A. Applications of Coarse-Grained Models in Metabolic Engineering. *Frontiers in Molecular Biosciences*. 2022; 9:806213. [https://www.frontiersin.org/articles/10.3389/fmolb.2022.806213](https://www.frontiersin.org/articles/10.3389/fmolb.2022.806213/full), doi: 10.3389/fmolb.2022.806213.
- Fraão CJR**, Trichez D, Serrano-Bataille H, Dagkesamanskaia A, Topham CM, Walther T, François JM. Construction of a synthetic pathway for the production of 1,3-propanediol from glucose. *Scientific Reports*. 2019; 9(1):11576. <https://www.nature.com/articles/s41598-019-48091-7>, doi: 10.1038/s41598-019-48091-7.
- Gerosa L**, Haverkorn van Rijsewijk BRB, Christodoulou D, Kochanowski K, Schmidt TSB, Noor E, Sauer U. Pseudo-transition Analysis Identifies the Key Regulators of Dynamic Metabolic Adaptations from Steady-State Data. *Cell Systems*. 2015; 1(4):270–282. <https://linkinghub.elsevier.com/retrieve/pii/S2405471215001465>, doi: 10.1016/j.cels.2015.09.008.
- Giordano N**, Mairet F, Gouzé JL, Geiselman J, De Jong H. Dynamical Allocation of Cellular Resources as an Optimal Control Problem: Novel Insights into Microbial Growth Strategies. *PLOS Computational Biology*. 2016; 12(3):e1004802. <https://dx.plos.org/10.1371/journal.pcbi.1004802>, doi: 10.1371/journal.pcbi.1004802.
- Izard J**, Gomez Balderas CD, Ropers D, Lacour S, Song X, Yang Y, Lindner AB, Geiselman J, De Jong H. A synthetic growth switch based on controlled expression of RNA polymerase. *Molecular Systems Biology*. 2015; 11(11):840. <https://www.embopress.org/doi/full/10.15252/msb.20156382>, doi: 10.15252/msb.20156382.
- Liang Q**, Zhang H, Li S, Qi Q. Construction of stress-induced metabolic pathway from glucose to 1,3-propanediol in *Escherichia coli*. *Applied Microbiology and Biotechnology*. 2011; 89(1):57–62. <https://link.springer.com/article/10.1007/s00253-010-2853-3>, doi: 10.1007/s00253-010-2853-3.
- Martin Y**, Page M, Blanchet C, De Jong H. WellInverter: a web application for the analysis of fluorescent reporter gene data. *BMC Bioinformatics*. 2019; 20(1):309. <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-2920-4>, doi: 10.1186/s12859-019-2920-4.
- Megazyme**. Glucose (GOPD format) Assay Procedure. Megazyme; 2023, https://prod-docs.megazyme.com/documents/Assay_Protocol/K-GLUC_DATA.pdf.
- Milo R**. What is the total number of protein molecules per cell volume? A call to rethink some published values. *BioEssays*. 2013; 35(12):1050–1055. <https://onlinelibrary.wiley.com/doi/10.1002/bies.201300066>, doi: 10.1002/bies.201300066.
- Pavlou A**, Cinquemani E, Pinel C, Giordano N, Van Melle-Gateau M, Mihalcescu I, Geiselman J, De Jong H. Single-cell data reveal heterogeneity of investment in ribosomes across a bacterial population. *Nature Communications*. 2025; 16(1):285. <https://www.nature.com/articles/s41467-024-55394-5>, doi: 10.1038/s41467-024-55394-5.
- Schroeder WL**, Suthers PF, Willis TC, Mooney EJ, Maranas CD. Current State, Challenges, and Opportunities in Genome-Scale Resource Allocation Models: A Mathematical Perspective. *Metabolites*. 2024; 14(7):365. <https://www.mdpi.com/2218-1989/14/7/365>, doi: 10.3390/metabo14070365.

- Scott M**, Gunderson CW, Mateescu EM, Zhang Z, Hwa T. Interdependence of cell growth and gene expression: Origins and consequences. *Science*. 2010; 330(6007):1099–1103. <https://www.science.org/doi/10.1126/science.1192588>, doi: 10.1126/science.1192588.
- Scott M**, Klumpp S, Mateescu EM, Hwa T. Emergence of robust growth laws from optimal regulation of ribosome synthesis. *Molecular Systems Biology*. 2014; 10(8):747. <https://link.springer.com/article/10.15252/msb.20145379>, doi: 10.15252/msb.20145379.
- Wong N**, Jantama K. Engineering *Escherichia coli* for a high yield of 1,3-propanediol near the theoretical maximum through chromosomal integration and gene deletion. *Applied Microbiology and Biotechnology*. 2022; 106(8):2937–2951. <https://link.springer.com/article/10.1007/s00253-022-11898-y>, doi: 10.1007/s00253-022-11898-y.
- Yabo AG**, Caillaud JB, Gouzé JL, De Jong H, Mairet F. Dynamical Analysis and Optimization of a Generalized Resource Allocation Model of Microbial Growth. *SIAM Journal on Applied Dynamical Systems*. 2022; 21(1):137–165. <https://epubs.siam.org/doi/10.1137/21M141097X>, doi: 10.1137/21M141097X.
- Yegorov I**, Mairet F, De Jong H, Gouzé JL. Optimal control of bacterial growth for the maximization of metabolite production. *Journal of Mathematical Biology*. 2019; 78(4):985–1032. <https://link.springer.com/10.1007/s00285-018-1299-6>, doi: 10.1007/s00285-018-1299-6.